

Bonsai 8

Two songs. Eight stems. Your mix.

User guide / Firmware 0.4.4 candidate and companion site / 3 October 2026

Bonsai 8 turns the SP-1 into a two-deck stem player. Choose a song for **Deck A** and another for **Deck B**. Bring their voices, drums and instruments together, shape each stem, and record the result on the player.

The four faders control one deck at a time. Tap **FUNCTION** to change decks. The other deck keeps playing with the levels you left it at.

Release status: The test player now runs the filter-only 0.4.4 candidate, with its four song entries unchanged. Eight-stem dry playback and USB audio alone passed short tests, but USB audio together with mirroring still caused dropouts. This is not a verified release. Check the exact version and test status in the website installer before updating. Opening the website does not update your player. The older 0.4.0–0.4.3 effect controls are noted in section 04.

Your first playback

1. Hold **FUNCTION** for about 1.5 seconds to power on. Both decks start paused; A is selected.
2. A loads the first occupied song slot; B loads the second. Start with the master volume low.
3. On A, move the faders to their maximum end to pick up the starting levels, then lower them to taste.
4. Tap and release **PLAY** to start A. Tap again to pause.
5. Tap **FUNCTION** to select B. Its levels start at zero: move the faders to minimum, then raise the stems you want.

At power-on, A's stem levels start at 100%, B's at 0%, master at 25%, and both speeds at normal. Songs remain stored; deck selections, mix settings and effects reset when you power off.

Keep this nearby

Website: sp-1.xyz

Alternate address: placid-shrine-2h5k.here.now

Illustrated guide: sp-1.xyz/guide.html

This guide covers mixing, physical controls, lights, effects, recording, the companion site, library management, installation and troubleshooting.

01 / Mix two songs

Use this example to put **Song 2's vocals over Song 1's instruments**. The standard stem order is 1 Vocals, 2 Drums, 3 Bass, 4 Other. Imported files must follow that order for these labels to match their sound.

A carries the instruments

1. Select A with a brief FUNCTION tap if needed. Side light 1 shows A is selected.
2. Hold FUNCTION and tap the side wheel to browse to Song 1. Release both. The chosen song is cued at the beginning, paused.
3. Mute Track 1, the vocal: tap its track button if it is not already muted.
4. Pick up faders 2, 3 and 4 and set the instrumental balance. Tap PLAY.

B brings in the vocal

5. Tap FUNCTION to select B. Side light 4 shows B is selected; A keeps playing.
6. Hold FUNCTION and tap the wheel to find Song 2. Release both.
7. Bring all B faders to minimum to pick up their initial zero levels. Leave 2, 3 and 4 at zero, or mute them. Raise fader 1 and make sure Track 1 is unmuted.
8. Tap PLAY at the point where you want the vocal to enter. Adjust fader 1 to balance it over A.

To try a different vocal, stay on B and browse to any other occupied slot with FUNCTION + wheel. B stops and cues the new song; A keeps its playback state. Tap PLAY to bring B back in. Empty slots are skipped and browsing wraps at the end. Both decks can also use the same song.

Make the timing work

Bonsai 8 does not automatically match beats or musical keys. Launch the second song by ear. Its four stems share one timeline, but A and B have separate timelines.

Move the wheel without FUNCTION to change the selected deck's speed and pitch together. With only B's vocal audible, this changes the vocal while A stays unchanged. The vocal's timing changes too: there is no separate pitch control for an individual stem or pitch-preserving stretch.

Songs loop at their full length. Hold PLAY for about half a second to cue the selected song at its beginning. To cue both, do this on A and B separately. Then FUNCTION + PLAY starts both together; it does not align their beats or intros.

02 / Physical controls

A **tap** is a short press and release. PLAY and track-button holds become a different action after about half a second. For a FUNCTION combination, press FUNCTION first, then the other control promptly; holding FUNCTION alone for 1.5 seconds powers off.

Control	Action
FUNCTION, tap	Select Deck A or Deck B.
FUNCTION, hold 1.5 seconds	Power on or power off.
PLAY, tap	Play or pause the selected deck.
PLAY, hold about 0.5 seconds	Cue the selected song to the beginning, paused.
FUNCTION + PLAY	Start both if either is paused. Pause both if both are playing.
A stem's fader	Adjust its volume on the selected deck, after pickup.
A track button, tap	Mute or unmute that stem on the selected deck.
A track button, hold about 0.5 seconds	Solo that stem on the selected deck until released.
Wheel, without FUNCTION	Change the selected deck's speed and pitch together.
FUNCTION + wheel	Browse all occupied songs on the selected deck.
Volume + / -	Adjust the combined master volume.
FUNCTION + a track button, held	Edit that stem's filter amount with its fader. The wheel has no effect while editing.
FUNCTION + both volume buttons, hold 1 second	Start or stop recording the mix. Release before repeating.
Track 1 + Track 4, hold 3 seconds	Enter bootloader mode for a firmware update.

Fader pickup

Each deck remembers its four levels during the session. After switching decks, move a fader through that deck's stored level before it takes control. This prevents a sudden jump in the mix. An unmuted track's light blinks while pickup is waiting.

If you do not know the stored level, sweep slowly to one end and back. Once pickup occurs, further movement changes the sound. The same rule applies when a fader returns from effect control to volume.

Solo affects only the selected deck. It does not silence the other deck or raise a fader that is at zero. Song changes keep the deck's current levels, mutes, speed and effects until power-off.

03 / Read the lights

The four small side status lights show which deck you control and which decks are playing. Count them in order along the side; the separate dark opening is not a status light.

Side light in normal playback	Meaning
1	Deck A is selected.
2	Deck A is set to play.
3	Deck B is set to play.
4	Deck B is selected.

Both middle lights on means both decks are set to play. A playing light is not a sound meter: muted stems or zero levels can still make that deck silent.

Track lights

A lit track light means that stem exists and is unmuted on the selected deck. It can stay lit while the song is paused or its fader is at zero. A blinking unmuted track light means its fader is waiting for pickup. Missing or muted stems have no light.

When the display changes

Situation	Side-light display
Hold FUNCTION to browse	Song-slot position and group. See below.
Edit a filter (0.4.4)	Side light 1 identifies Filter.
Recording	All four blink slowly.
Recording stopped; draining or saving	All four blink faster. Pause both decks and keep power on.
Recording saved	Normal deck lights return.
Recording failed	The outer lights flash. Do not assume the take was saved.
Storage unavailable or unsupported	Alternating pairs. Do not erase the library to troubleshoot this.

During a file transfer, all four track lights blink. Recording indicators take priority over the effect and deck display.

Song-slot display

While FUNCTION is held, a steady side light gives the slot's position within a group of four. A blinking light gives its group: light 1 for slots 1-4, light 2 for 5-8, light 3 for 9-12, and light 4 for 13-16.

For example, slot 6 shows light 2 steady and light 2 blinking at the same position, so the steady light hides that blink. Use the connected site's A/B readouts when you need an unambiguous song number.

04 / Shape a stem

In **0.4.4**, each of the eight deck/stem positions has its own **high-pass filter**. Increasing its amount removes more bass and low frequencies while keeping the brighter highs. The filter stays active after you release the controls, including when that deck changes songs. Its setting resets at power-off.

1. Tap FUNCTION to select the deck you want.
2. Hold FUNCTION and that stem's track button together. Keep both held while editing. Side light 1 identifies Filter.
3. Move that stem's fader through its stored filter amount to pick it up, then adjust. After power-on, begin at minimum to pick up the starting amount of zero.
4. Release the buttons. The fader returns to volume control with pickup, so it will not jump the stem's level.

To bypass the filter, set its amount to zero using the same gesture. The wheel has no effect while editing the filter. Release the buttons to use the wheel for its normal deck speed/pitch control.

Older firmware, 0.4.0–0.4.3: These versions have Filter, Echo and Reverb. While holding FUNCTION + a stem button, the wheel cycles those three choices; side lights 1, 2 and 3 identify them. That stem's fader sets the amount, and zero bypasses it. This is the historical mapping. The installed 0.4.4 removes Echo and Reverb and uses the filter-only controls above. Removing them alone does not establish that USB playback is reliable.

Volume and pitch while mixing

Use the selected deck's four faders to balance its stems. The volume buttons change the combined master output. There is no separate deck crossfader: build fades with the stem levels.

The wheel changes the selected deck's speed from **0.5x to 1.25x**. Pitch follows speed, and all audible stems on that deck change together. To pitch just the vocal against another song's drums, isolate the vocal on one deck and put the drums on the other.

Begin with moderate stem levels. If the combined mix distorts, lower the stem levels and master. If an effect itself sounds distorted, reduce its amount or bypass it. Turning the master down cannot undo distortion already created inside an effect.

Headphones carry the same combined mix as the speaker; there is no separate headphone preview deck.

05 / Record your mix

Record A and B directly on the SP-1, without a computer. The take includes the audible mix, effects, pitch changes and master-volume moves. It saves to the **first empty song slot**; at least one of the 16 slots must be free.

Start, perform, stop

1. Prepare your mix and start playback.
2. Hold FUNCTION and both volume buttons for **one second**, then release. All four side lights blink slowly while recording.
3. Perform with the faders, mutes, deck switch, wheel and effects. Song changes and library transfers are unavailable until the take is finished and saved.
4. Hold FUNCTION and both volume buttons for one second again, then release to stop. The side lights blink faster as the take finishes.

Pause both decks to save

5. Pause both decks. If both are playing, FUNCTION + PLAY pauses them together. If only one is playing, select it and tap PLAY. **Do not use FUNCTION + PLAY in that case: it would start both.**
6. Keep the player powered on while it saves. Muting both songs is not enough; both transports must be paused. Wait for the normal deck lights to return before powering off, unplugging or entering bootloader mode.
7. Browse to the new song or find it in **On your player** on the site. Play it using Track 1, unmuted, with its fader raised. Export it later from the site if you want a WAV file.

The take is one stereo mix on **Track 1**. Tracks 2-4 are empty; recording does not split the finished mix back into eight stems. It may replay more quietly through the player's normal mix levels. Export preserves the recorded level.

Limits to remember

A take lasts up to about eight minutes. At the limit it stops automatically, then follows the same pause-and-save process. The take is not ready until saving succeeds.

Flashing outer side lights indicate failure. Check the site's library before treating a take as saved. If storage cannot keep up or verification fails, the failed take is not published as a playable song.

Keep power connected during saving. Switching off or losing power during a storage write can lose the take or damage its library entry. Existing songs are not intentionally overwritten.

06 / Use the companion site

Open sp-1.xyz. The site is a browser instrument and a companion to the physical player. You can prepare music without hardware, or connect the SP-1 to follow its controls, lights and playback.

Make stems in the browser

1. Choose **Drop a song here** and select a WAV, MP3, M4A or FLAC file.
2. Choose **Separate into 4 stems**. Keep the tab open while it works. The first run downloads a large separation model; processing can take several minutes.
3. Use the browser's play button, faders, Mute and Solo to explore the song. Speed / pitch changes both together.
4. Choose **Export stems** to save the prepared stems as a ZIP of WAV files.

Already have stems? Choose **Have stems?** and assign your files to Vocals, Drums, Bass and Other. Use a common starting point and keep any leading silence so they stay aligned. Missing stems stay silent. The studio demo is another quick way to try the browser mixer.

Mix two browser songs

Choose Deck A or Deck B before importing a song. The four stem controls apply to the selected deck. Use each deck's song selector to load any song kept in this tab, its Play button to start or pause it, and its Deck level to balance it against the other song. Cue returns that deck to the beginning. Changing one deck leaves the other playing.

Try both demos loads two generated songs for practice. **Play both** starts both; **Pause both** stops both. Remove an unused song from the session to free memory. The session library is temporary and disappears when the page is reloaded, so export work you want to keep.

On a keyboard, A and B select a deck, Space plays or pauses it, Shift + Space controls both, and keys 1-4 mute its stems. Shortcuts do not override typing in a field and are inactive while the physical player is connected.

Separation runs on your device. Audio is not uploaded to a server. Browser files must be under 200 MB each and at most ten minutes; songs sent to the physical player have the shorter limit on the next page. Export stems saves the prepared source stems, not a recording of your browser mix.

The file-size limit bounds decoding and separation memory: a compressed song expands into audio samples, four output stems and processing buffers. The browser mixer also limits retained decoded audio to 768 MiB. That budget does not include every temporary allocation or the separation model. Use shorter files and remove unused session songs if memory is exhausted.

Match tempo, key and individual stem pitch

1. Load songs on both browser decks and open **Match songs**. Choose the master deck whose tempo and key you want to keep.
2. Choose **Analyze both songs**. Check the BPM and first-beat positions. Use Half or Double if the estimate follows the wrong pulse, or enter a corrected value. Key can remain Unknown; do not treat an estimate as certain.
3. Keep **Match the other deck's tempo to the master** enabled to prepare the follower at the master's tempo. Enable key matching only after checking both keys. It shifts vocals, bass and other; drums retain their own pitch setting.
4. Set each follower stem's semitone offset. For example, +2 raises only its chosen stem by two semitones while preserving the prepared song's duration. These offsets add to any key-matching shift.
5. Choose **Prepare Deck A** or **Prepare Deck B**. The finished song is selected for export/upload, and its original remains in the tab's library. Cancel leaves the existing songs intact.
6. Choose **Align on next beat** to start or align the two grids on the browser audio clock. Listen and correct the grids if needed. Export the selected prepared stems, or connect the player and upload them to an empty slot.

This is browser preparation, not live automatic matching on a disconnected SP-1. Prepared files can play on the physical player afterward, but physical starts still need manual timing. A fixed beat grid cannot follow changing tempo. Pitch processing can soften transients or change vocal tone; it does not preserve formants or turn a major song into a minor arrangement. The normal speed/pitch rocker still changes timing and pitch together.

Connect the player

1. Use Chrome or Edge on a computer for USB features. Connect with a USB data cable and power on normally. Close any other app using the player's connection.
2. Choose **Connect SP-1**, keep **Bonsai 8 / Dual Deck** selected, then **Connect player**. Select Bonsai 8 in the chooser. Older compatible firmware may use the name SP-1 Dual Deck.
3. Use the physical controls. The digital model follows reported fader positions, button presses and lights. The connected site shows the selected deck's settings and both song positions.
4. Choose **Listen here** to hear the physical stereo mix through the computer and see its waveform. Allow audio-input permission; the site uses the player's named USB input.

The connected 3D model is a monitor. Its browser controls do not remotely move physical faders or change physical pitch. **Load A / Load B** in the library are separate actions that can choose the songs on the player.

USB audio is the combined mix, not eight separate input channels. The physical master changes its level. Use computer output volume for local listening. Listening through both outputs can sound like an echo because USB audio and the screen add delay. The song timers show source position, not beat markers.

07 / Manage songs

Connect Bonsai 8, then use **On your player**. It lists occupied song slots and their lengths. A successful site upload keeps its source title for that connection. Current firmware does not store original filenames or a durable content identity, so reconnecting can show Song 1, Song 2 and so on. The site does not guess names from reused slots.

Choose any song for either deck

Use **Load A** or **Load B** beside a song. That deck stops and cues the chosen song; the other deck keeps its playback state. Select the target deck on the physical player and tap PLAY when you want it to enter. Refresh reloads the library list.

Upload prepared stems

1. Separate a song in the browser, or import existing stems with **Have stems?**
2. Connect the player and choose an empty destination under **Save prepared stems to**.
3. Choose **Upload these stems**. Keep USB connected and the tab open until the site confirms that upload and verification finished.
4. Use Load A or Load B to cue the new song, then press PLAY on that deck.

The player has **16 song slots**, with up to four stems in each. Uploads are limited to about eight minutes. Use matching starting points; shorter stems are padded with silence. The player itself does not separate a mixed song into instruments.

Upload sends the selected prepared stems, including completed Match songs tempo and per-stem pitch processing. It does not include temporary browser fader, mute, solo, deck-level or tape-speed settings. Keep a copy of the original stems on your computer.

Export or delete

Choose **Export** beside a song to download it. A single-stem song, including a recorded mix, exports as a WAV. A multistem song exports as a ZIP of WAVs.

Choose **Delete** beside the song you want to remove and check the slot in the confirmation. Export it first if you want a copy. Deletion frees that slot for a new upload; it does not securely erase the old audio data.

Uploading, exporting and deleting pause both decks. These operations stop USB listening too. Wait for completion, then restart physical playback and choose Listen here again if needed.

Library actions are blocked while recording, draining or waiting to save. Stop recording and pause both decks first. Do not disconnect during a transfer or start a second uploader. A firmware update does not require you to upload your existing songs again.

08 / Install or update Bonsai 8

The website's **Install Bonsai 8** button updates the application on the player. It does not intentionally erase or replace the music library. Read the version and release status shown in the installer before continuing.

1. Finish any recording or file transfer. Disconnect the site from the player and close other player connections.
2. Keep USB connected. On the powered-on SP-1, hold **Track 1 + Track 4 for three seconds** to enter bootloader mode. Playback stops.
3. Open **Install Bonsai 8** on sp-1.xyz. Wait for the published firmware check to finish.
4. Select the installation checkbox, then choose **Choose bootloader & install**. Select the SP-1 bootloader in the browser's device chooser.
5. Keep USB connected and the tab open until the installer reports completion. Do not start a library transfer during the update.
6. Hold FUNCTION for about 1.5 seconds to power on if needed. Reconnect normally, check the reported version and song list, and try playback and the controls.

The installer checks the downloaded image before sending it and checks the bootloader's responses. Successful transfer still needs a normal power-on and playback check afterward.

Which mode should I use?

Normal power-on is for music playback, live mirroring, USB listening and managing songs. **Bootloader mode** is only for installing firmware. The live controls and audio are not available there.

If the chooser finds no bootloader, close it, enter bootloader mode again and retry. If USB connection options are unavailable, open the site in Chrome or Edge on a computer. A cable that only charges cannot transfer firmware or audio.

About Bonsai 8

Bonsai 8 is custom community firmware for the SP-1, based on the SP-1 Tape Looper project. It is not an official Teenage Engineering release. The companion site includes source and license credits in **About this project**.

For illustrated, step-by-step help, open sp-1.xyz/guide.html.

09 / If something feels wrong

What you notice	Try this
Deck B plays silently	B starts with all levels at zero. Select B, move its faders to minimum for pickup, raise the wanted stems, unmute them and check master.
A fader does nothing	Cross its stored level to pick it up. Switching decks or leaving effect mode can require pickup again.
PLAY restarts the song	The press lasted long enough to cue. Use a short tap for pause/resume.
FUNCTION powers off	Hold it alone for less than 1.5 seconds when switching decks; press the second control promptly for combinations.
Solo still leaves another song audible	Solo affects only the selected deck. Pause or mute the other deck separately.
Songs drift or clash	Match their launch and speed by ear. There is no automatic beat or key synchronization; speed changes pitch too.
The effect seems stuck on	Hold FUNCTION + that track button and pick up its effect amount, then move the fader to zero.
Recording stopped but no new song appears	Pause both decks and wait for normal side lights. Outer flashing lights mean the take failed. Check for an empty slot before trying again.
A recorded mix only uses Track 1	Correct: the complete stereo mix is on Track 1. The other three stems are empty.
The site cannot connect	Power on normally, use a data cable and desktop Chrome or Edge, and close other player connections.
No sound through Listen here	Check audio-input permission and whether the computer sees a Bonsai 8 or SP-1 Dual Deck USB input. Reconnect the powered-on player.
Music becomes choppy	Stop Listen here and bypass the filter to compare. The earlier 0.4.2 failed the eight-stem USB/mirroring test with two filters; installed 0.4.4 still needs its own hardware tests. A moving timer does not prove clean audio.
Side lights alternate in pairs	Storage is unavailable or unrecognized. Reconnect and check the site status; do not erase an existing library as a troubleshooting step.

For a repeatable problem, note the firmware version, songs on A and B, active stems, speeds, effects, and whether recording or USB listening was active. Keep your source audio backed up.